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Recursive VAR Identification of Risk-Induced Exchange Rate Dynamics

Evidence from the Yen Carry Trade

Galedo, Enrique Lorenzo · Go, Keira · Cuenca, Raphael · Patajo, Juliana · Sison, Aaron Joshua E.
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INTRODUCTION

Why This Study?

The Motivating Event

- **March 31, 2026**: Japan's Finance Minister Satsuki Katayama branded the yen's sharp decline as "speculative" for the first time since the Iran war began, and signaled Tokyo's readiness to intervene "on all fronts" (Reuters, 2026).
- Escalating Middle East tensions (Iran war, Strait of Hormuz disruption) ignited a **triple sell-off** in Japanese assets:
  - Yen plunged to near ¥160/USD
  - Nikkei 225 lost over 11% in March
  - 10-year JGB yield rose to levels unseen since 1999
- **Structural break in carry trade profitability:**
  - Yen was the world's premier funding currency for decades under near-zero QQE rates
  - Strait of Hormuz disruption drove oil prices up, pushing imported inflation into Japan
  - BOJ forced toward rate hikes, and markets priced ~70% probability of April 2026 hike
  - Rising rates compressed the interest differential, triggering a self-reinforcing unwind
  - Investors repatriated yen, sold foreign assets, and dumped JGBs to 1999 levels
  - Funding currency cheapness was endogenously destroyed, reversing the carry trade incentive

Source: Reuters (2026, March 31). "Japan brands yen falls as 'speculative' as Iran war ignites sell-off."

Why It Is Significant

- This crisis is a **natural experiment** in risk transmission, and the yen's behavior during acute geopolitical stress violates standard UIRP predictions.
- Risk-off dynamics drove yen *appreciation* through carry trade unwinding and safe-haven flows, exposing the **forward premium puzzle** in real time.
- The event creates an **exogenous shock** to VIX, oil prices, and FX risk reversals, which are three key risk indicators, and this allows us to trace their causal impact on JPY/USD while interest rate differentials remained relatively stable.

CAUSAL CLAIM

Causal Relationship & Research Question

> *A one-standard-deviation positive shock to financial risk indicators (VIX or FX risk reversals) causes a statistically significant appreciation of the Japanese Yen against the US Dollar within 2–3 days, with effects persisting over a 7-day horizon.*
>
> Proposed causal claim

Two Indirect Transmission Channels

Carry Trade Unwinding

Leveraged investors liquidate yen-funded positions, purchasing yen to repay liabilities during global uncertainty spikes.

Safe-Haven Rebalancing

Portfolio shifts toward yen-denominated assets, consistent with Uncovered Equity Parity (Hau & Rey, 2006).

Research Question

What is the causal effect of exogenous shocks to financial risk indicators (VIX, oil price, and FX risk reversals) on the JPY/USD exchange rate, operating through yen-funded carry trade unwinding and safe-haven demand channels, after controlling for interest rate differentials, FX liquidity, and relative equity performance?

Base Article for Methodology

Guyot, O., Montgomery, H. A., & Yang, P. (2026). Risk Premiums, Market Volatility, and Exchange Rate Dynamics: Evidence from the Yen Carry Trade. *Risks, 14*(2), 46.

- **Method adopted:** Reduced-form VAR with Cholesky decomposition
- **Key finding replicated:** Risk indicators (VIX, risk reversals) and equity gap Granger-cause JPY/USD returns; interest rate differentials do not
- **Our extension:** Extended sample to include the 2025–2026 Iran crisis period and substituted Bloomberg Terminal data with public APIs (FRED, Yahoo Finance, MOF Japan, BIS, Iacoviello GPR)

RESEARCH DESIGN

The Ideal Experiment

An ideal experiment establishes clear cause-and-effect relationships by scientifically manipulating variables while controlling external factors. **Three core pillars:**

1. Random Assignment

Randomly assign trading days to a "treatment" group that receives an exogenous risk shock (e.g., a sudden VIX spike) and a "control" group that does not. Randomization ensures observed differences in JPY/USD returns are attributable to the shock, not pre-existing group differences.

2. Control Group

A baseline of observations where no risk shock occurs, measuring the *counterfactual*, or what JPY/USD would have done absent the shock. Compare "normal" days (low VIX, stable oil, narrow risk reversals) against "shock" days (high VIX, oil spike, wide risk reversals).

3. Manipulation of the IV

The researcher *exogenously* varies the risk indicator (e.g., VIX level) and observes the resulting change in the dependent variable (JPY/USD return), holding all other variables constant (interest rate differentials, liquidity, equity gap).

Why infeasible. We cannot *randomly assign* geopolitical risk events or manipulate VIX levels in real financial markets. Exchange rates are determined by simultaneous interactions across thousands of market participants. This infeasibility motivates our **identification strategy**, which uses observational data and econometric assumptions to approximate the ideal experiment.

IDENTIFICATION STRATEGY

Recursive (Cholesky) VAR Identification

An identification strategy is the specific set of assumptions and research designs used to isolate and measure a true causal effect from observational data. Based on **Guyot, Montgomery & Yang (2026)**.

POS.	VARIABLE	ECONOMIC JUSTIFICATION	AFFECTS ↓ / AFFECTED BY ↑
1	$\Delta YCurve^{Diff}$	Monetary policy changes slowly; does not react to same-day FX (Eichenbaum & Evans, 1995)	Affects 2–8; affected by none
2	$\Delta(i - i*)$	Central banks meet infrequently; same-day FX cannot move policy rates	Affects 3–8; affected by 1
3	ΔBrent	Global oil prices are exogenous to JPY; Japan is a price taker	Affects 4–8; affected by 1–2
4	VIX	US equity fear index reflects global risk; not driven by JPY liquidity	Affects 5–8; affected by 1–3
5	Spread	FX liquidity responds to global risk, not vice versa	Affects 6–8; affected by 1–4
6	RiskReversal	FX hedging demand responds to risk and liquidity	Affects 7–8; affected by 1–5
7	EGAP	Equity return gap responds to risk, then drives FX via portfolio rebalancing (Hau & Rey, 2006)	Affects 8; affected by 1–6
8	ΔJPY/USD	Most endogenous, and the exchange rate absorbs all contemporaneous shocks	Affects none; affected by 1–7

Key Identification Restrictions

- Monetary policy variables (positions 1–2) are **contemporaneously exogenous** to same-day FX
- Global risk factors (positions 3–4) affect FX but are not caused by FX within the same day
- FX microstructure variables (positions 5–6) respond to risk but do not cause global risk
- The exchange rate (position 8) is the **residual clearing price**, and it absorbs all contemporaneous shocks

Novel Extension, the GPR Subsample

- Iacoviello's daily Geopolitical Risk (GPR) index classifies trading days as **"High Stress"** (GPR > 90th percentile) vs. "Normal"
- Enables **subsample VAR estimation** testing for nonlinear regime-dependent transmission
- Addresses Brunnermeier et al.'s (2009) VIX threshold nonlinearity finding

Causal Inference Tests

Granger causality · Orthogonalized impulse response functions (IRFs) · Forecast error variance decomposition (FEVD)

DATA

Data Sources & Variable Construction

#	VARIABLE	PROXY / SERIES CODE	SOURCE	RETRIEVAL
1	ΔJPY/USD	DEXJPUS, USDJPY=X	FRED, Yahoo	FRED API, Yahoo Finance API
2	$\Delta(i - i*)$	1Y JGB (MOF), DFF (FRED)	MOF Japan, FRED	MOF website, FRED API
3	$\Delta YCurve^{Diff}$	JGB 10Y/2Y; DGS10/DGS2	MOF Japan, FRED	MOF website, FRED API
4	ΔBrent	DCOILWTICO (WTI), BZ=F (Brent)	FRED, Yahoo	FRED API, Yahoo Finance API
5	VIX	CBOE Volatility Index (^VIX)	Yahoo Finance	Yahoo Finance API
6	Spread	(High − Low) / Close proxy	Yahoo Finance	Yahoo Finance API
7	RiskReversal	1M JPY/USD 25Δ Risk Reversal	BIS (quarterly)	stats.bis.org
8	EGAP	Nikkei 225 (^N225), S&P 500 (^GSPC)	Yahoo Finance	Yahoo Finance API
9	GPR	Geopolitical Risk Index (daily)	Iacoviello (2026)	matteoiacoviello.com

a **Data Substitution:** Original Guyot et al. (2026) used Bloomberg Terminal data. Daily Japanese short-term bank rates unavailable from public APIs (FRED publishes monthly only); substitute 1Y JGB yield from MOF Japan. Spread pending OANDA API access; uses intraday range proxy.

b **Data Limitation:** 1-month JPY/USD 25-delta risk reversal published only at quarterly frequency by BIS. Interpolated to daily via cubic spline.

Sample period: Jan 1, 2018 to May 31, 2026 (~1,800 to 2,000 trading days after NYSE/TSE alignment). All series from free, publicly accessible sources, and no Bloomberg Terminal is required.

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